

Fundamentals of Computer graphics

Academic year 2017/2018 - 2^o Semester

Individual Practical Work

With this practical work is intended that students use the software called Blender to create a short **film** and a small **game**. All work should be based on the knowledge taught in class, it may, however, be used knowledge obtained elsewhere.

– Film –

The creation of the film should be based on the knowledge taught in the practical classes. More specifically, should meet the specifications listed below. Accordingly, when:

- **Modelling:**

- It will only be evaluated models created from scratch, i.e., imported models will have no score;
- The constructed models should use different primitive's types, like:
 - *Mesh* (for example, *Cube*);
 - *Curves* (for example, *Bezier Curve*);
 - *Surfaces* (for example, *Nurbs Sphere*);
 - 3D Text.
- The modelling process must be implemented with the following techniques:
 - Change of primitive parameters;
 - Extrusions;
 - Modelling based in reference images (the images must be part of the delivered files and without them this technique cannot be evaluated);
 - Modifiers (it's mandatory to use at least three (3) modifiers that specifically help the modeling task).

- **Shading:**

- Demonstrate, necessarily, the influence of at least 3 different types of light source in the created scenes;
- The created models must have different types of materials, and it is required materials:
 - Where the brightness is treated (it is mandatory to have at least two (2) materials with different values for the brightness);
 - Transparent (it is mandatory to have at least two (2) materials with different values for the transparency);
 - Reflective;

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- Based on procedural textures (is required to use at least two (2) different types of this texture, for example, *Clouds* and *Marble*);
 - Based on image textures (is required to use at least two (2) different images).
 - In the application of image-based textures it must be used the following:
 - UV unwrapping technique;
 - Bump Mapping technique;
 - Application of textures to non-planar shapes such as tubes, cubes or balls, without using UV unwrapping technique (is required to apply at least to two (2) objects of different shape).
 - **Animation:**
 - Must have different ways of using animation, necessarily, each of the following techniques:
 - Keyframing;
 - Armatures composed of a minimum of three (3) bones, each bone being responsible for a movement distinct from the others;
 - Combination of Shapekeys (implies the existence of two (2) different Shapekeys, used together, in the same model);
 - Physics (the use of different physical simulations will be valorised);
 - Particle system;
 - Modifiers (it is mandatory to use at least one (1) that specifically helps the model animation task, other than the armature modifier).
 - **Rendering:**
 - The film should have a minimum duration of 45 seconds, where all the above content is presented;
 - Generate a film with the right size and codec for a good visualization;
 - Failure to render the film or non-compliance with the requested duration will imply the grade zero (0) to the film component;
 - It should be incorporated in the movie necessary at least one (1) sound file. This incorporation must be made only with Blender.
- Game –
- The game to be implemented may contain new graphics or use those that have been created and used in the movie and must,
- Having a screen menu;

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- Have, at least, one (1) animation.
- Handle keyboard and mouse events;
- Have, at least, one (1) “intelligent” element;
- Have, at least, two (2) elements with different types of physics;
- Treat, at least, two (2) collisions made by different ways among different elements;
- Using, at least, two (2) game properties;
- Using, at least, one (1) text game property;
- Use, at least, two (2) different cameras, inside the same game’s scene;
- Have, at least, one (1) script;
- Have an executable version.

Evaluation

This work worth 13 values of the final grade, divided as follows:

- Film - 10 Values;
- Game - 3 Values.

Each practical work must be defended before the discipline’s professors and its final grade will depend of the demonstration of knowledge that the student shows on the defense, in addition of what it was made by him. So, in summary, the evaluation will consider the following aspects:

- The implementation of demanded content and mentioned above;
- Demonstration of knowledge that the student presents in the defense (will be valued the knowledge of techniques and effects that were not taught in class);
- The quality and originality of animation and game.

The student must have 35% of minimum grade in each part of the practical work (film and game), without which the student will not be subject to approval.

Extra Information

- Practical work should be done individually;
- Any fraud detected will lead to failing the students involved;
- Practical work must be submitted in digital format, in a date to be announced. At that time, the student should deliver within a folder labelled with his name the following:
 - Blender’s projects;
 - The film and the game created;
 - Everything that has been used in the implementation of this practical work (external models, images used in textures, sounds, etc.).
- Once delivered, it is not possible to improve the grade of this practical work.